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DIPLOMA THESIS

Assessment of Physical Rehabilitation Exercises using Deep Learning and Computer Vision

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ABSTRACT

Physical rehabilitation is essential for helping patients regain functional mobility, yet its success is heavily bottlenecked by a lack of expert supervision in home environments. While automated motion assessment systems offer a promising and scalable solution, traditional models often operate as “black boxes,” struggling with interpretability and failing to account for the natural variability between different patients’ bodies.

This thesis introduces KinetoCheck, an AI-assisted platform designed to assess movement using standard RGB and RGB-D cameras. The system leverages a two-stream deep learning framework: one stream uses Spatio-Temporal Graph Attention Networks (STGAT) to analyze spatial body topology, while a 1D-CNN stream tracks joint angle behavior over time. Trained on the University of Idaho Physical Rehabilitation Movement Data (UI-PRMD), the model relies on triplet margin loss and Batch-Hard mining to learn a reliable, subject-invariant standard for movement quality. Moving beyond simple pass/fail binary classification, KinetoCheck provides patients with a continuous quality score (ranging from 0 to 1). Crucially, it uses graph attention weights to visually identify “Joints of Attention” (JoA), giving both clinicians and patients clear, explainable feedback on specific biomechanical errors.

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Chapter 1

Introduction

1.1 Motivation

Physical rehabilitation is critical for restoring mobility after surgery, injury, or stroke. However, most routines are completed at home without clinician supervision, which can lead to incorrect movements, poor muscle engagement, and secondary injuries. Advances in accessible sensing technologies, such as RGB-based pose tracking and affordable depth cameras, now enable the development of automated virtual coaching systems for home rehabilitation. This thesis aims to provide anatomically meaningful guidance that identifies where movement quality declines, moving beyond basic correct or incorrect feedback.

1.2 Problem Statement

Automated rehabilitation assessment differs fundamentally from generic action recognition. Recognition predicts what action is being performed, while rehabilitation assessment must estimate how well that action is performed. Most existing approaches simplify this challenge to binary classification, which fails to capture gradual patient progress. In addition, models trained on relatively small rehabilitation datasets are prone to identity leakage, where subject-specific body characteristics are learned instead of movement quality patterns. This limits generalization, especially under strict Leave-One-Subject-Out (LOSO) evaluation.

1.3 Aims and Objectives

The primary aim of this thesis is to design an interpretable and subject-invariant framework for rehabilitation exercise quality assessment.

The objectives are:

- implement geometric normalization strategies that reduce subject-specific bias due to body proportions;
- design a two-stream STGAT architecture that fuses coordinate-based and clinically meaningful angle-based representations;
- use metric learning with triplet loss to model movement quality on a continuous scale in the interval $[0, 1]$;
- extract and visualize graph attention patterns as Joints of Attention (JoA) to provide localized corrective feedback.

1.4 Thesis Structure

- Chapter 2 provides the theoretical background on biomechanics, pose representations, graph-based learning, and metric learning for quality assessment.
- Chapter 3 reviews related work, tracing the evolution from traditional temporal alignment techniques to modern graph-neural approaches for interpretable rehabilitation feedback.
- Chapter 4 details the proposed methodology and the KinetoCheck pipeline, including preprocessing, two-stream feature construction, and fusion for continuous scoring.
- Chapter 5 presents the training design, with emphasis on Batch-Hard triplet mining, subject-invariant optimization, and LOSO evaluation.
- Chapter 6 describes the implementation stack and the real-time inference pipeline for RGB and depth-based data sources.
- Chapter 7 reports quantitative and qualitative results, ablation studies, and error analysis.
- Chapter 8 concludes the thesis, summarizes contributions, and discusses future research directions.

Chapter 2

Theoretical Background

This chapter provides the mathematical and anatomical foundations for modeling rehabilitation exercises as spatio-temporal graphs.

2.1 Biomechanics and Rehabilitation Kinematics

Physical rehabilitation assessment differs from general activity recognition because it requires detecting subtle deviations within the same motion class.

In action recognition, an algorithm mainly answers what action is being performed. In rehabilitation analysis, the main question is how well that action is executed. Two repetitions may belong to the same exercise class and still differ clinically: one may preserve joint alignment and controlled range of motion, while the other contains compensatory behavior. This distinction motivates quality-centered modeling rather than category-centered modeling.

For KinetoCheck, this means the input representation must preserve clinically meaningful joint behavior over time. In practical terms, trunk stability, joint flexion trajectory, and inter-limb symmetry are treated as first-class signals, not secondary artifacts.

2.1.1 Clinical Indicators

Assessment relies on range of motion (ROM) and specific joint flexion/extension angles. These indicators are central for identifying whether a patient executes a movement with correct biomechanical alignment.

From a kinematic perspective, the same endpoint posture can still hide incorrect execution if the path contains abrupt compensatory motions. Therefore, ROM and angle peaks are interpreted together with temporal smoothness. In KinetoCheck, both positional trajectories and angle trajectories are modeled so that local joint mechanics and full-body context remain coupled during inference.

If three landmarks A , B , and C define a joint centered at B , the corresponding angle can be computed from vectors:

$$\theta = \arccos \left(\frac{(A - B) \cdot (C - B)}{\|A - B\| \|C - B\|} \right) \quad (2.1)$$

This formulation is used for clinically interpretable descriptors in the two-stream pipeline.

2.1.2 Performance Taxonomy

Movements are categorized into Optimal (standard execution by a professional) and Non-Optimal (compensatory or incorrect patterns). This taxonomy supports both model training and clinically interpretable feedback.

In project practice, this taxonomy acts as a supervision anchor for both classification and metric learning. Optimal samples define reference behavior, while Non-Optimal samples model deviation classes that include reduced ROM, poor posture control, and timing inconsistency. This binary structure is simple but clinically useful, and it also supports extension to continuous quality scores by embedding-based distances.

2.2 Skeletal Data Representation

2.2.1 Graph Formulation

The human body is modeled as a non-Euclidean graph $G = (V, E)$, where V represents joints (nodes) and E represents anatomical connections (edges). This representation preserves topology that would be lost in conventional grid-based formulations.

Given a sequence of T frames and N joints, the input tensor can be written as:

$$X \in \mathbb{R}^{T \times N \times C} \quad (2.2)$$

where C is the coordinate channel dimension (typically 2D or 3D). The graph structure is encoded by an adjacency matrix $A \in \mathbb{R}^{N \times N}$ and, in STGAT variants, by learned attention matrices that modulate neighborhood importance per frame.

KinetoCheck uses this abstraction across different acquisition backends (dataset skeletons, MediaPipe landmarks, and real-time sensor streams), enabling one conceptual model family despite heterogeneous data sources.

2.2.2 Normalization

To ensure subject invariance, coordinates are root-centered (typically at the pelvis) and scaled by torso length. This reduces biases related to subject height and camera distance.

This step is critical in the project because training and inference inputs can come from different capture systems. Root-centering removes translation variance, while torso-based scaling attenuates body-size and depth-scale differences. Let p_i^t be joint i at time t , p_{root}^t the pelvis/root joint, and s^t a torso-length scale factor. The normalized joint is:

$$\tilde{p}_i^t = \frac{p_i^t - p_{root}^t}{s^t + \epsilon} \quad (2.3)$$

with small $\epsilon > 0$ for numerical stability.

KinetoCheck also uses joint mapping steps (for example 32-joint sensor output to a UI-PRMD-compatible skeleton subset) so that inference-time representations remain compatible with trained models.

2.3 Spatio-Temporal Graph Attention Networks (STGAT)

STGAT extends graph convolutional networks by adding dynamic importance weighting across joints and edges.

The motivation for attention is straightforward: not all joints contribute equally at every phase of an exercise. During a squat-like movement, lower-limb joints may dominate quality cues; during upper-limb rehabilitation, shoulder-elbow-wrist chains may carry most discriminative signal.

2.3.1 Graph Attention Mechanism

Unlike fixed convolutions, STGAT uses self-attention to adaptively determine connection strengths between neighboring joints based on the specific phase of an exercise.

If h_i and h_j are node features for joints i and j , attention coefficients are computed from learned projections and normalized over neighborhoods. This allows the model to increase focus on clinically relevant joints only when needed. In KinetoCheck-related experiments, these weights are further inspected as interpretability artifacts (Joints of Attention) rather than used only as latent internals.

2.3.2 Multi-Head Attention

Multiple attention heads allow the model to learn distinct movement characteristics in parallel, such as postural stability and trajectory consistency.

From a modeling perspective, multi-head attention improves representational diversity and reduces dependence on a single attention pattern. Different heads can specialize in complementary subspaces, such as local articulation versus global body coordination. This is particularly useful in rehabilitation, where quality deviations may be subtle and distributed across multiple joints.

2.4 The Temporal Pyramid Principle

The Temporal Pyramid structure is a multi-scale modeling technique used to capture motion dynamics across different time resolutions.

In the KinetoCheck backend branch dedicated to temporal pyramid STGAT, this principle is used to improve robustness against variable execution speed and frame-rate differences across videos.

2.4.1 Multi-Scale Sampling

Motion sequences are subsampled at varying rates (for example 1/1, 1/2, and 1/4) and processed by parallel sub-networks.

Each branch processes a different temporal granularity. Fine-rate streams preserve rapid local changes, while coarse-rate streams capture long-range trends and phase structure. Branch outputs are fused into a shared representation that balances sensitivity and temporal context.

2.4.2 Capturing Speed Variations

This architecture handles the variable speeds at which patients perform exercises, extracting meaningful features from both slow recovery movements and faster repetitions.

This property is practical in home environments where cadence is inconsistent and unsupervised. Temporal pyramids reduce sensitivity to absolute speed and prioritize relative movement quality, which better aligns with rehabilitation progress tracking.

2.5 Deep Metric Learning and Triplet Loss

The framework uses deep metric learning to map exercise executions into a latent quality space.

Beyond hard class boundaries, embedding-space learning supports graded similarity. This is useful when patient movements progressively approach target quality over multiple sessions.

2.5.1 Triplet Margin Loss

The objective is to minimize the distance between an anchor (correct) and a positive (correct) sample while maximizing the distance from a negative (incorrect) sample.

For embeddings $f(x)$, a common formulation is:

$$\mathcal{L}_{triplet} = \max(0, d(f(x_a), f(x_p)) - d(f(x_a), f(x_n)) + m) \quad (2.4)$$

where m is a margin and $d(\cdot, \cdot)$ is a distance metric, usually Euclidean distance on normalized embeddings.

2.5.2 Batch-Hard Mining

Batch-Hard mining selects the most difficult triplets within each mini-batch, which provides stronger gradients and helps prevent training stagnation.

This strategy is especially relevant for medium-size rehabilitation datasets, where random triplets quickly become uninformative. Selecting hardest positives and hardest negatives increases gradient utility per batch and improves convergence stability.

In KinetoCheck development, this idea is coupled with subject-aware evaluation protocols (for example LOSO) to reduce identity leakage and improve true cross-subject generalization.

2.6 Project-Level Theoretical Integration in KinetoCheck

The theoretical components above are not isolated modules; they are integrated into an end-to-end workflow spanning data preprocessing, model training, API serving, and real-time inference.

2.6.1 Two-Stream Perspective

The experimental two-stream branch combines STGAT processing over joint trajectories with temporal modeling over angle signals. The first stream preserves topological motion context, while the second stream injects direct clinical semantics. Their fusion aligns with the core thesis hypothesis that coordinate and angle modalities are complementary.

2.6.2 Deployment-Oriented Constraints

The backend architecture introduces practical constraints that influence model theory choices: robustness to variable video quality, support for different pose extractors, compatibility across skeleton formats, and predictable inference latency. Consequently, normalization and representation consistency are treated as theoretical requirements, not just implementation details.

2.6.3 From Theory to Explainable Feedback

Attention-enabled graph modeling provides a direct path from mathematical representation to user-facing feedback. By aggregating attention patterns into joint-level importance cues, the system can indicate where quality degradation occurs, supporting clinician interpretation and patient correction during home-based rehabilitation.

Chapter 3

Related Work

This chapter reviews the evolution of automated rehabilitation assessment, from traditional algorithms to modern two-stream graph-based approaches.

3.1 Traditional Movement Assessment

Early systems relied on non-learning-based techniques, including Dynamic Time Warping (DTW) and Gaussian Mixture Models (GMM). While these methods were interpretable, they struggled with subject variability and lacked the capacity to learn deep hierarchical movement representations.

3.2 The Pioneer Deep Learning Framework (Liao et al.)

The first significant deep-learning implementation for rehabilitation assessment was proposed by Liao et al. (2020).

3.2.1 Architecture

The model introduced a Temporal Pyramid + LSTM design, where parallel 1D-CNN branches processed joint displacement signals at different time scales before temporal integration with a multi-layer LSTM.

3.2.2 Significance

This work established UI-PRMD as a benchmark dataset and demonstrated that hierarchical spatio-temporal modeling is essential for movement quality regression.

3.3 Contemporary GCN and Transformer Frameworks

Recent research has moved toward graph-based architectures that preserve human body topology more effectively.

3.3.1 ExeChecker (2024–2025)

ExeChecker introduced the concept of Joints of Attention (JoA). By extracting attention weights from an STGAT backbone, it provides localized feedback on which joints contributed most to an incorrect classification.

3.3.2 D-STGCNT

D-STGCNT combines dense graph convolutions with GRU and Transformer components to produce high-accuracy quality scores while maintaining efficient inference.

3.3.3 EGCN++

EGCN++ represents a strong state-of-the-art direction on UI-PRMD by fusing position and orientation (angle) features, reporting very high classification performance.

3.4 Video-Based and Smartphone Telerehabilitation

Recent implementations by Chander et al. (2025) transitioned these methods to RGB-only settings using MediaPipe keypoint extraction. Their results highlight the feasibility of home deployment through live repetition counting via autocorrelation and lightweight Transformer-based recognition.

Chapter 4

Proposed Methodology

4.1 Data Acquisition and Preprocessing

4.1.1 UI-PRMD Data Pipeline

4.2 Two-Stream Feature Engineering (Positions and Angles)

4.2.1 Coordinate and Angle Complementarity

4.3 STGAT Backbone for Spatial Topology Modeling

4.3.1 Attention-Driven Graph Encoding

4.4 Recurrent Integration for Long-Term Temporal Dependencies

4.4.1 Temporal Context Aggregation

4.5 Fused Latent Space and Continuous Scoring

4.5.1 Quality Manifold Construction

Chapter 5

Training and Optimization Strategies

5.1 Triplet Network Architecture

5.1.1 Embedding Space Design

5.2 Implementation of Batch-Hard Triplet Mining

5.2.1 Gradient Informativeness in Small Batches

5.3 Subject-Invariant Normalization and Augmentation

5.3.1 Identity Leakage Mitigation

5.4 Leave-One-Subject-Out Cross-Validation Protocol

5.4.1 Strict Generalization Evaluation

Chapter 6

Implementation and System Integration

6.1 Video-Based Inference Pipeline via MediaPipe

Chapter 7

Results and Discussion

7.1 Classification and Regression Performance Analysis

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